Position-dependent motion gain with an unknown wall position

1 Real-time estimation with the wall position as a state

Motion gain as a function of the distance from the wall

$$
\begin{aligned}
h[k] &= \bar{x}_d[k] - \delta\bar{x}_3[k] - \bar{w}_s \\
\bar{a}'[k] &= \frac{1-c}{h[k]^2} \\
M[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\bar{x}_3[k]
\end{aligned}
$$

State of the process

$$
\begin{aligned}
\delta\bar{x}_1[k+1] &= \delta\bar{x}_2[k] \\
\delta\bar{x}_2[k+1] &= \delta\bar{x}_3[k] \\
\delta\bar{x}_3[k+1] &= \lambda_c\,\delta\bar{x}_3[k] - F_{dx}[k]\,e_{\bar{a}}[k] - \varepsilon_x[k] \\
\bar{a}[k+1] &= \bar{a}[k] + \bar{a}'[k]\left(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\right) \\
c[k+1] &= c[k] \\
\bar{w}_s[k+1] &= \bar{w}_s[k]
\end{aligned}
$$

Measurement and output matrix

$$
\mathbf{y}[k] = \begin{bmatrix} \delta\bar{x}_m[k] \\ \bar{a}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{x}}_1[k] \\ \hat{\bar{a}}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}
$$

Estimator

$$
\begin{aligned}
\hat{h}[k] &= \bar{x}_d[k] - \delta\hat{\bar{x}}_3[k] - \hat{\bar{w}}_s[k] \\
\hat{\bar{a}}'[k] &= \frac{1-\hat{c}[k]}{\hat{h}[k]^2} \\
\hat{M}[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]
\end{aligned}
$$

$$
\begin{aligned}
\delta\hat{\bar{x}}_1[k+1] &= \delta\hat{\bar{x}}_2[k] + \left\{\ell_{11}[k]\,e_{x_1}[k] + \ell_{12}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_2[k+1] &= \delta\hat{\bar{x}}_3[k] + \left\{\ell_{21}[k]\,e_{x_1}[k] + \ell_{22}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{x}}_3[k] + \left\{\ell_{31}[k]\,e_{x_1}[k] + \ell_{32}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{a}}[k+1] &= \hat{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\hat{M}[k] + \left\{\ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]\right\} \\
\hat{c}[k+1] &= \hat{c}[k] + \left\{\ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{w}}_s[k+1] &= \hat{\bar{w}}_s[k] + \left\{\ell_{61}[k]\,e_{x_1}[k] + \ell_{62}[k]\,e_{a_m}[k]\right\}
\end{aligned}
$$

Dynamics of estimation errors

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{x}_1}[k] \\ e_{\delta\bar{x}_2}[k] \\ e_{\delta\bar{x}_3}[k] \\ e_{\bar{a}}[k] \\ e_{c}[k] \\ e_{\bar{w}_s}[k] \end{bmatrix}, \qquad \mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & 0 & 0 \\ 0 & 0 & \hat{\bar{a}}'[k]\,(1-\lambda_c) + \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] & 1 + \hat{\bar{a}}'[k]\,F_{dx}[k] & -\dfrac{\hat{M}[k]}{\hat{h}[k]^2} & \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_x[k] \\ \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ 0 \\ 0 \end{bmatrix}$$

Row 4 of $\mathbf{F}_e$, expanded

(a) the two gain steps and their difference

$$\bar{a}[k+1] - \bar{a}[k] = \bar{a}'[k]\,\big(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\big)$$
$$\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\,\hat{M}[k]$$
$$e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = \bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] + \bar{a}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \bar{a}'[k]\,\varepsilon_x[k]$$

(b) each factor written as its estimate plus its error

$$h[k] = \hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k], \qquad c = \hat{c}[k] + e_c[k]$$
$$\bar{a}'[k] = \frac{1 - \hat{c}[k] - e_c[k]}{\big(\hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k]\big)^2} = \hat{\bar{a}}'[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big) - \frac{1}{\hat{h}[k]^2}\,e_c[k] + O(e^2)$$
$$M[k] = \hat{M}[k] + (1 - \lambda_c)\,e_{\delta\bar{x}_3}[k]$$

(c) the product difference, exact

$$\bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] = \hat{\bar{a}}'[k]\,\big(M[k] - \hat{M}[k]\big) + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\,\hat{M}[k] + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\big(M[k] - \hat{M}[k]\big)$$

(d) every term after substituting (b)

$$\begin{aligned} e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = {} & \hat{\bar{a}}'[k]\,(1 - \lambda_c)\,e_{\delta\bar{x}_3}[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\delta\bar{x}_3}[k] - \frac{\hat{M}[k]}{\hat{h}[k]^2}\,e_c[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ & + \Big(\frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big) - \frac{e_c[k]}{\hat{h}[k]^2}\Big) \\ & \qquad \times \Big((1 - \lambda_c)\,e_{\delta\bar{x}_3}[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\Big) + O(e^2) \end{aligned}$$

(e) first-order terms collected by error

$$\begin{aligned} e_{\bar{a}}[k+1] = {} & \Big(\hat{\bar{a}}'[k]\,(1 - \lambda_c) + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\Big)\,e_{\delta\bar{x}_3}[k] \\ & + \big(1 + \hat{\bar{a}}'[k]\,F_{dx}[k]\big)\,e_{\bar{a}}[k] \\ & - \frac{\hat{M}[k]}{\hat{h}[k]^2}\,e_c[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \end{aligned}$$

Process noise and measurement noise

$$\mathbf{Q}[k] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & q_{33}[k] & -\hat{\bar{a}}'[k]\, q_{33}[k] & 0 & 0 \\ 0 & 0 & -\hat{\bar{a}}'[k]\, q_{33}[k] & \hat{\bar{a}}'[k]^2\, q_{33}[k] & 0 & 0 \\ 0 & 0 & 0 & 0 & q_{55}[k] & 0 \\ 0 & 0 & 0 & 0 & 0 & q_{66} \end{bmatrix}$$

$$q_{55}[k] = P_{\text{fix}}^2\, I[k], \qquad I[k] = \frac{1}{p_{55}[k]} - \frac{1}{p_{55}^f[k]}$$

$$q_{66} = 0$$

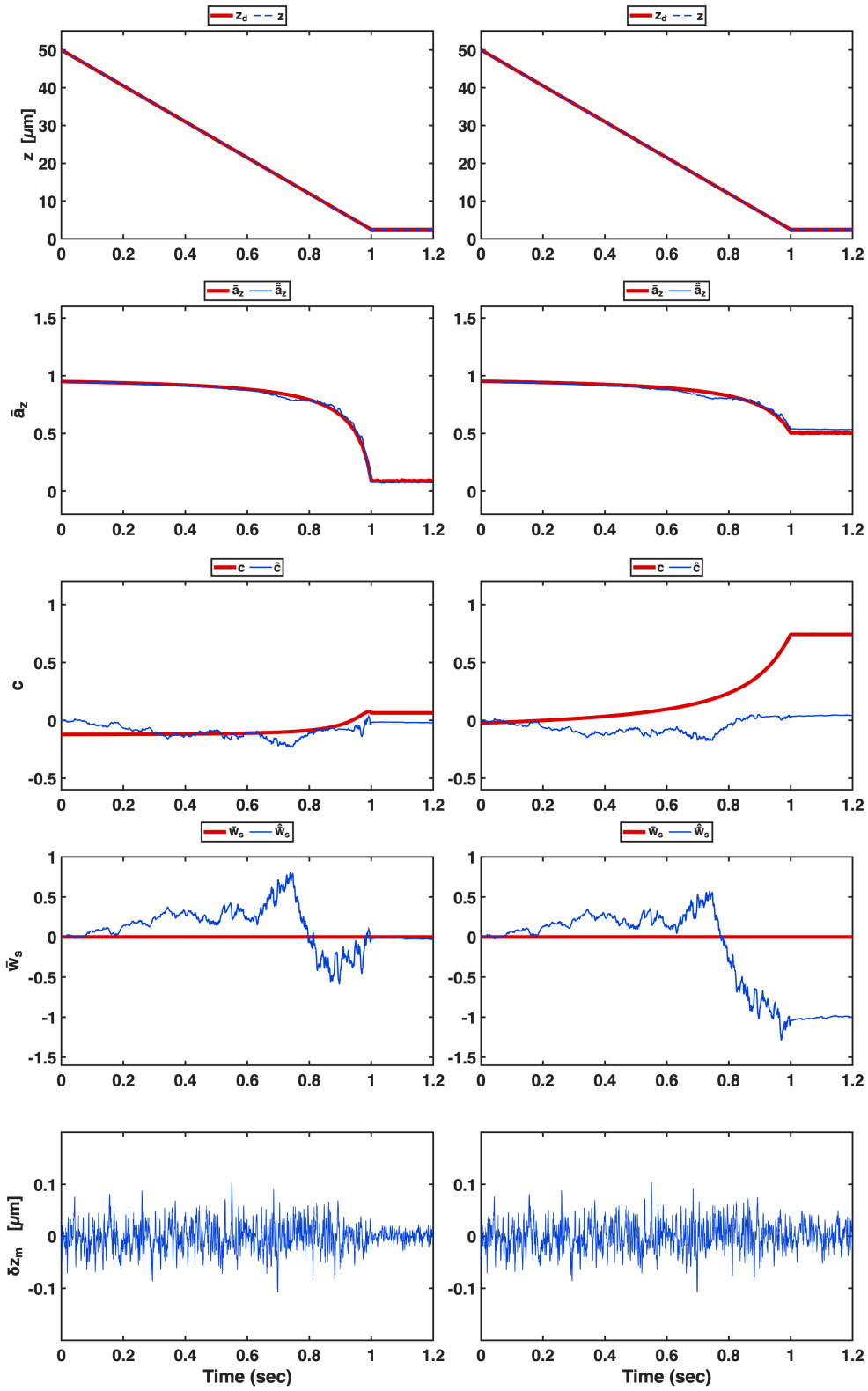
$$\mathbf{R}[k] = \operatorname{diag}\big(r_1,\ r_2[k]\big)$$

(q_{33}, r_1, r_2, $\mathbf{L}$, $\mathbf{P}$, $\mathbf{P}^f$ as in motion_ctrl Section 4; q_{55} as in motion_ctrl Section 6.)

2 Constant-velocity descent

Fig. 2.1 Descent 22.2 → 1.10 R

z axis; 22.2 → 1.10 R at constant velocity in 1 s, then parked at 1.10 R for 0.2 s; left: plane wall; right: cell wall; red: $c(h_d) = 1 - h_d^2\,\bar{a}'(h_d)$ and $\bar{w}_s = 0$; $\sqrt{P_{\text{fix}}} = 0.15$, $q_{66} = 0$, $\hat{c}[0] = 0$, $\hat{\bar{w}}_s[0] = 0$, $\sqrt{p_{66}[0]} = 0.64$; $\lambda_c = 0.4$, seed 1



Error propagation (no noise, coefficients at the estimate)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k]$$

$$\mathbf{e}[k_0+j] = \mathbf{\Phi}(k_0+j,k_0)\,\mathbf{e}[k_0], \qquad \mathbf{\Phi}(k_0+j,k_0) = \mathbf{F}_e[k_0+j-1]\cdots\mathbf{F}_e[k_0], \qquad \mathbf{\Phi}(k_0,k_0) = \mathbf{I}$$

Errors at step k_0, output errors j steps later

$$\begin{bmatrix} e_{x_1}[k_0+j] \\ e_{a_m}[k_0+j] \end{bmatrix} = \mathbf{H}\,\mathbf{\Phi}(k_0+j,k_0)\,\mathbf{e}[k_0]$$

Observability matrix from step k_0 to step k_1

$$\begin{bmatrix} e_{x_1}[k_0] \\ e_{a_m}[k_0] \\ e_{x_1}[k_0+1] \\ e_{a_m}[k_0+1] \\ \vdots \\ e_{x_1}[k_1] \\ e_{a_m}[k_1] \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{H} \\ \mathbf{H}\,\mathbf{\Phi}(k_0+1,k_0) \\ \vdots \\ \mathbf{H}\,\mathbf{\Phi}(k_1,k_0) \end{bmatrix}}_{\mathbf{O}(k_0,k_1)} \begin{bmatrix} e_{\delta\bar{x}_1}[k_0] \\ e_{\delta\bar{x}_2}[k_0] \\ e_{\delta\bar{x}_3}[k_0] \\ e_{\bar{a}}[k_0] \\ e_c[k_0] \\ e_{\bar{w}_s}[k_0] \end{bmatrix}$$

Window of three steps (6×6): $k_1 = k_0+2$, the window moves one step at a time

$$\mathbf{O}(k_0,\,k_0+2) = \begin{bmatrix} \mathbf{H} \\ \mathbf{H}\,\mathbf{F}_e[k_0] \\ \mathbf{H}\,\mathbf{F}_e[k_0+1]\,\mathbf{F}_e[k_0] \end{bmatrix}, \qquad \mathbf{O}(k_0+1,\,k_0+3) = \begin{bmatrix} \mathbf{H} \\ \mathbf{H}\,\mathbf{F}_e[k_0+1] \\ \mathbf{H}\,\mathbf{F}_e[k_0+2]\,\mathbf{F}_e[k_0+1] \end{bmatrix}$$

Window from the start of the descent: $k_0 = 0$, the window grows one step at a time

$$\mathbf{O}(0,\,k) = \begin{bmatrix} \mathbf{H} \\ \mathbf{H}\,\mathbf{\Phi}(1,0) \\ \vdots \\ \mathbf{H}\,\mathbf{\Phi}(k,0) \end{bmatrix}, \qquad \mathbf{O}(0,\,k+1) = \begin{bmatrix} \mathbf{O}(0,\,k) \\ \mathbf{H}\,\mathbf{\Phi}(k+1,0) \end{bmatrix}, \qquad \mathbf{\Phi}(k+1,0) = \mathbf{F}_e[k]\,\mathbf{\Phi}(k,0)$$

Fig. 2.2 Three-step window $\mathbf{O}(k_0,\, k_0+2)$ along the descent: σ_4, σ_5, σ_6 and $\mathbf{v}_4$, $\mathbf{v}_5$, $\mathbf{v}_6$

z axis; run of Fig. 2.1; one window at every step k_0, plotted at $t = k_0 T_s$; σ_i, $\mathbf{v}_i$ = singular values and right singular vectors of $\mathbf{O}(k_0,\, k_0+2)$; rows: h_d, σ_4–σ_6, the $\bar{a}$, c, $\bar{w}_s$ components of $\mathbf{v}_4$, $\mathbf{v}_5$, $\mathbf{v}_6$ (sign of each $\mathbf{v}_i$ fixed by a positive $v_{4,\bar{a}}$, $v_{5,c}$, $v_{6,\bar{w}_s}$); left: plane wall; right: cell wall; seed 1

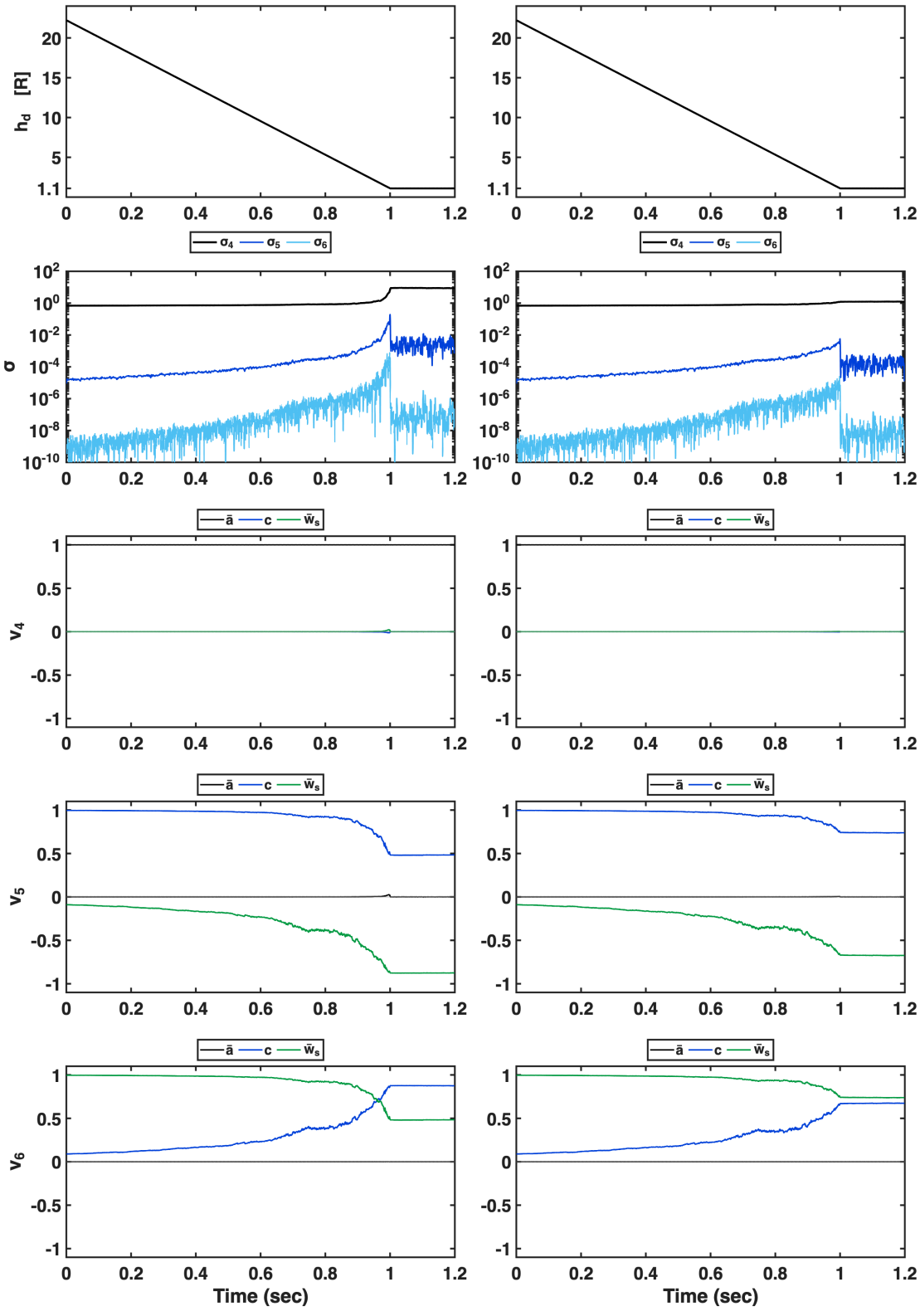
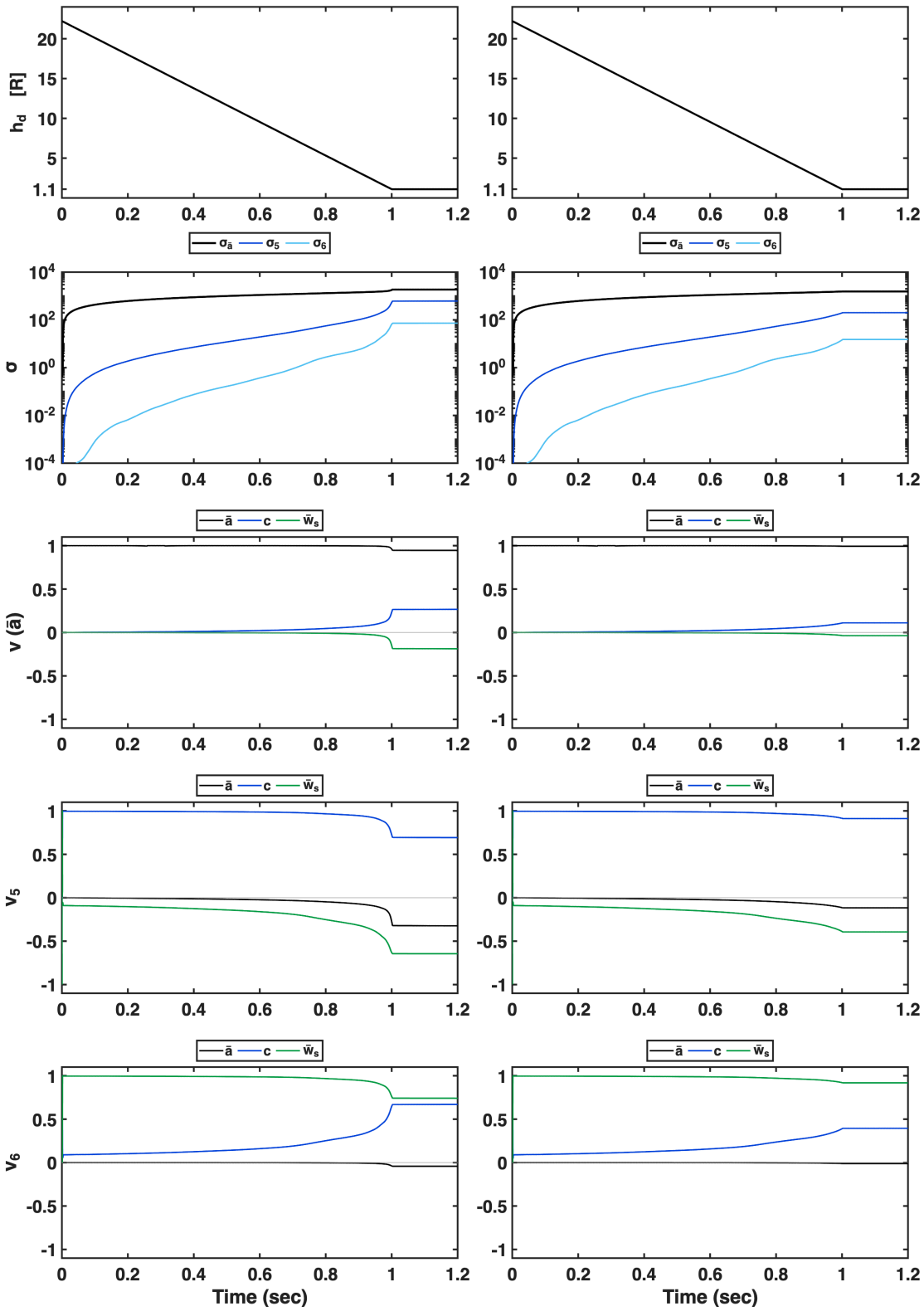


Fig. 2.3 Window from the start $\mathbf{O}(0,\, k)$ along the descent: σ and $\mathbf{v}$ of $\bar{a}$, σ_5, σ_6, $\mathbf{v}_5$, $\mathbf{v}_6$

z axis; run of Fig. 2.1; $\mathbf{O}(0,\, k)$ at every step k, plotted at $t = kT_s$; $\mathbf{v}(\bar{a})$ = the right singular vector with the largest $\bar{a}$ component (the largest σ once the $\bar{a}_m$ readings accumulate), $\mathbf{v}_5$, $\mathbf{v}_6$ = the two with the largest c, $\bar{w}_s$ components; rows: h_d, the three σ, the $\bar{a}$, c, $\bar{w}_s$ components of $\mathbf{v}(\bar{a})$, $\mathbf{v}_5$, $\mathbf{v}_6$; left: plane wall; right: cell wall; seed 1



3 c-only with a wall-position error

Wall of the plant at $\bar{w}_s = \Delta$ (plane wall; plant only)

$$D_\perp(u) = 1 - \tfrac{9}{8}\,u + \tfrac{1}{2}\,u^3 - \tfrac{57}{100}\,u^4 + \tfrac{1}{5}\,u^5 + \tfrac{7}{200}\,u^{11} - \tfrac{1}{25}\,u^{12}$$

$$\bar{a}_{\text{plant}}(h) = D_\perp\Big(\frac{1}{h-\Delta}\Big)$$

Estimator (motion_ctrl Section 6, unchanged: wall at $\bar{w}_s = 0$)

$$\hat{h}[k] = \bar{x}_d[k] - \delta\hat{\bar{x}}_3[k]$$

$$\hat{\bar{a}}'[k] = \frac{1 - \hat{c}[k]}{\hat{h}[k]^2}$$

$$q_{55}[k] = P_{\text{fix}}^2\, I[k]$$

Value of c of the wall at the true gap $g = h - \Delta$

$$\bar{a}'_{\text{plant}}(h) = \frac{1 - c(g)}{g^2} \quad \Longrightarrow \quad c(g) = 1 - g^2\,\bar{a}'_{\text{plant}}(h), \qquad g = h - \Delta$$

Trajectory

$h_d:\ 22.2 \ \rightarrow\ 1.10 + \max(\Delta, 0)$ in 1 s, then hold 0.2 s

Fig. 3.1 Wall farther than assumed: $\Delta = 0,\ -2,\ -4$ R

z axis; c-only of motion_ctrl Section 6, $\sqrt{P_{\text{fix}}} = 0.15$ in every case; red in the c row: $c(g)$ at the true gap of the particle, dotted: numerical clamp of $\hat{c}$, $[-3, 1]$; $\hat{c}[0] = 0$, $\lambda_c = 0.4$, seed 1

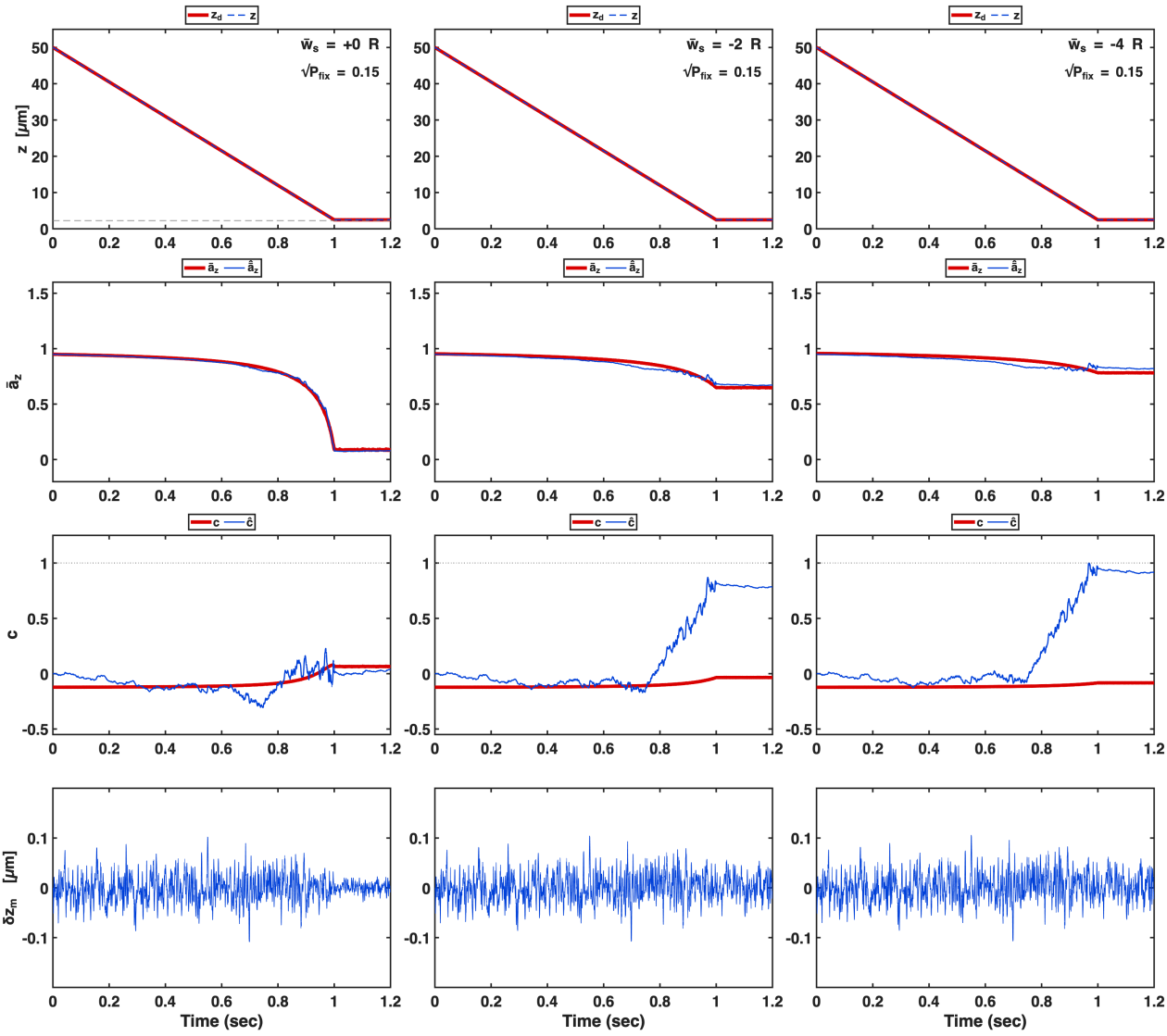
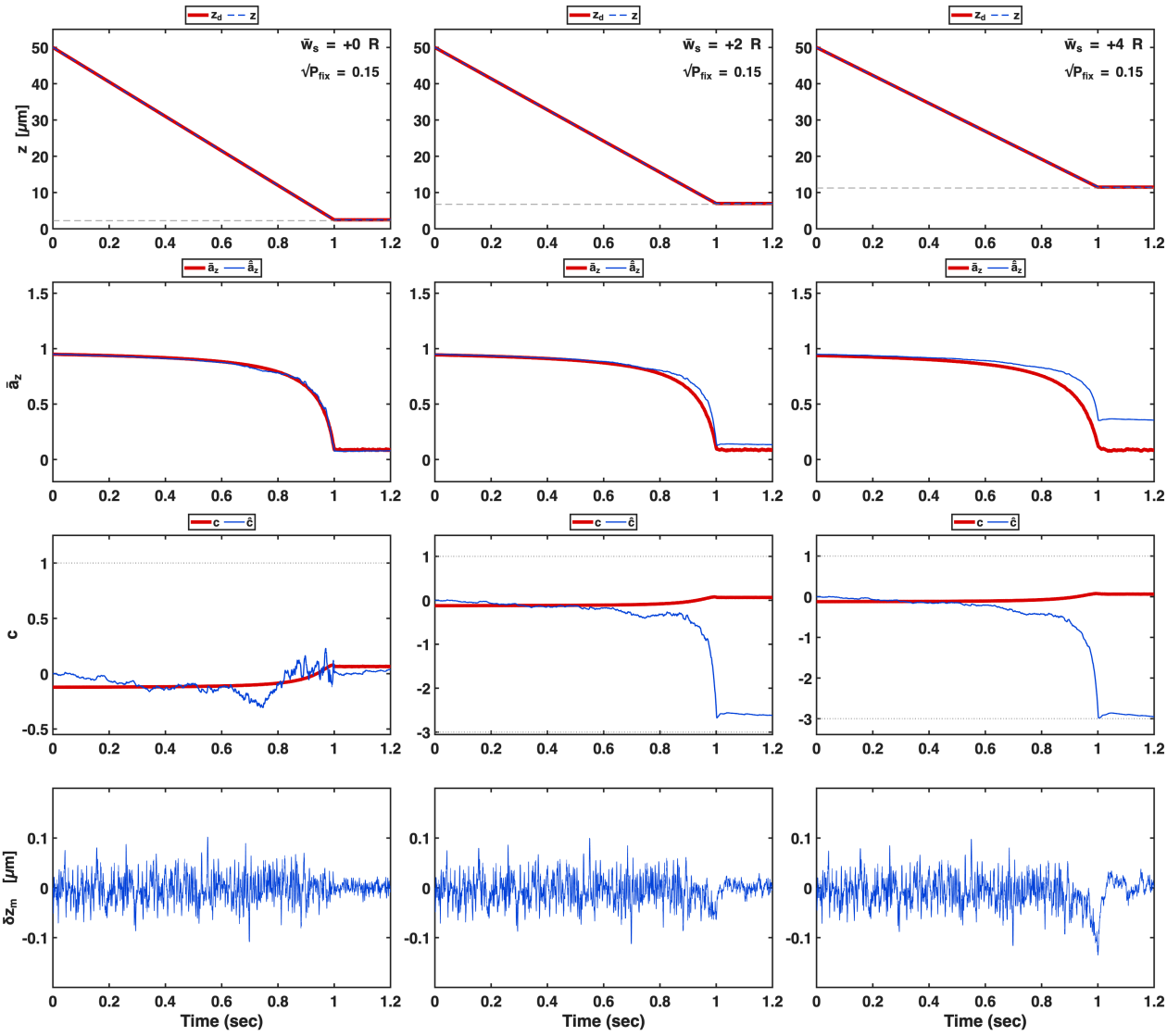


Fig. 3.2 Wall closer than assumed: $\Delta = 0,\ +2,\ +4$ R

z axis; bottom $h_d = 1.10 + \Delta$ (true gap 1.10); c-only of motion_ctrl Section 6, $\sqrt{P_{\text{fix}}} = 0.15$ in every case; red in the c row: $c(g)$ at the true gap of the particle, dotted: numerical clamp of $\hat{c}$, $[-3, 1]$; $\hat{c}[0] = 0$, $\lambda_c = 0.4$, seed 1



4 Slope with unit numerator, the wall position as the state

Motion gain as a function of the distance from the wall

$$
\begin{aligned}
h[k] &= \bar{x}_d[k] - \delta\bar{x}_3[k] - \bar{w}_s \\
\bar{a}'[k] &= \frac{1}{h[k]^2} \\
M[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\bar{x}_3[k]
\end{aligned}
$$

State of the process

$$
\begin{aligned}
\delta\bar{x}_1[k+1] &= \delta\bar{x}_2[k] \\
\delta\bar{x}_2[k+1] &= \delta\bar{x}_3[k] \\
\delta\bar{x}_3[k+1] &= \lambda_c\,\delta\bar{x}_3[k] - F_{dx}[k]\,e_{\bar{a}}[k] - \varepsilon_x[k] \\
\bar{a}[k+1] &= \bar{a}[k] + \bar{a}'[k]\left(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\right) \\
\bar{w}_s[k+1] &= \bar{w}_s[k]
\end{aligned}
$$

Measurement and output matrix

$$
\mathbf{y}[k] = \begin{bmatrix} \delta\bar{x}_m[k] \\ \bar{a}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{x}}_1[k] \\ \hat{\bar{a}}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}
$$

Estimator

$$
\begin{aligned}
\hat{h}[k] &= \bar{x}_d[k] - \delta\hat{\bar{x}}_3[k] - \hat{\bar{w}}_s[k] \\
\hat{\bar{a}}'[k] &= \frac{1}{\hat{h}[k]^2} \\
\hat{M}[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]
\end{aligned}
$$

$$
\begin{aligned}
\delta\hat{\bar{x}}_1[k+1] &= \delta\hat{\bar{x}}_2[k] + \left\{\ell_{11}[k]\,e_{x_1}[k] + \ell_{12}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_2[k+1] &= \delta\hat{\bar{x}}_3[k] + \left\{\ell_{21}[k]\,e_{x_1}[k] + \ell_{22}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{x}}_3[k] + \left\{\ell_{31}[k]\,e_{x_1}[k] + \ell_{32}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{a}}[k+1] &= \hat{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\hat{M}[k] + \left\{\ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{w}}_s[k+1] &= \hat{\bar{w}}_s[k] + \left\{\ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]\right\}
\end{aligned}
$$

Dynamics of estimation errors

$$
\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{x}_1}[k] \\ e_{\delta\bar{x}_2}[k] \\ e_{\delta\bar{x}_3}[k] \\ e_{\bar{a}}[k] \\ e_{\bar{w}_s}[k] \end{bmatrix}, \qquad \mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]
$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & 0 \\ 0 & 0 & \hat{\bar{a}}'[k]\,(1-\lambda_c) + \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] & 1 + \hat{\bar{a}}'[k]\,F_{dx}[k] & \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_x[k] \\ \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ 0 \end{bmatrix}$$

Row 4 of $\mathbf{F}_e$, expanded

(a) the two gain steps and their difference

$$\bar{a}[k+1] - \bar{a}[k] = \bar{a}'[k]\,\big(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\big)$$
$$\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\,\hat{M}[k]$$
$$e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = \bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] + \bar{a}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \bar{a}'[k]\,\varepsilon_x[k]$$

(b) each factor written as its estimate plus its error

$$h[k] = \hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k]$$
$$\bar{a}'[k] = \frac{1}{\big(\hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k]\big)^2} = \hat{\bar{a}}'[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big) + O(e^2)$$
$$M[k] = \hat{M}[k] + (1-\lambda_c)\,e_{\delta\bar{x}_3}[k]$$

(c) the product difference, exact

$$\bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] = \hat{\bar{a}}'[k]\,\big(M[k] - \hat{M}[k]\big) + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\,\hat{M}[k] + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\big(M[k] - \hat{M}[k]\big)$$

(d) every term after substituting (b)

$$\begin{aligned} e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = {} & \hat{\bar{a}}'[k]\,(1-\lambda_c)\,e_{\delta\bar{x}_3}[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\delta\bar{x}_3}[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big)\Big((1-\lambda_c)\,e_{\delta\bar{x}_3}[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\Big) + O(e^2) \end{aligned}$$

(e) first-order terms collected by error

$$\begin{aligned} e_{\bar{a}}[k+1] = {} & \Big(\hat{\bar{a}}'[k]\,(1-\lambda_c) + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\Big)\,e_{\delta\bar{x}_3}[k] \\ & + \big(1 + \hat{\bar{a}}'[k]\,F_{dx}[k]\big)\,e_{\bar{a}}[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \end{aligned}$$

Process noise and measurement noise

$$\mathbf{Q}[k] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & q_{33}[k] & -\hat{\bar{a}}'[k]\,q_{33}[k] & 0 \\ 0 & 0 & -\hat{\bar{a}}'[k]\,q_{33}[k] & \hat{\bar{a}}'[k]^2\,q_{33}[k] & 0 \\ 0 & 0 & 0 & 0 & q_{55} \end{bmatrix}, \qquad q_{55} = 0$$

$$\mathbf{R}[k] = \mathrm{diag}\big(r_1,\ r_2[k]\big)$$

(q_{33}, r_1, r_2, $\mathbf{L}$, $\mathbf{P}$, $\mathbf{P}^f$ as in motion_ctrl Section 4.)